

Tuning Systems Exercises

Catharina Hock, Stefanie Hövermann

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Exponential Functions and Logarithms

Exercise 1: Powers

Calculate the following.

1. $2^5 =$ _____
2. $10^3 =$ _____
3. $5^0 =$ _____
4. $3^4 =$ _____
5. $2^{-2} =$ _____

Exercise 2: Exponential Growth

A bacteria culture starts with 200 bacteria. Every hour the number doubles.

1. Complete the table.

Time (h)	0	1	2	3	4
Bacteria	200	_____	_____	_____	_____

2. Write a formula for the number of bacteria after t hours.

3. How many bacteria are there after 6 hours?

Exercise 3: Exponential Decay

A radioactive substance loses half of its mass every day. Initially it has 80 g.

1. Complete the table.

Days	0	1	2	3	4
Mass (g)	80	_____	_____	_____	_____

2. Write a formula for the mass after t days.

3. After how many days is only 10 g left?

Exercise 4: Evaluating Logarithms

Calculate.

1. $\log_2(8) = \underline{\hspace{2cm}}$
2. $\log_2(32) = \underline{\hspace{2cm}}$
3. $\log_{10}(1000) = \underline{\hspace{2cm}}$
4. $\log_5(25) = \underline{\hspace{2cm}}$
5. $\log_3(81) = \underline{\hspace{2cm}}$

Exercise 5: Rewrite as Exponentials

Rewrite each logarithm as an exponential equation.

1. $\log_2(16) = 4$
2. $\log_5(125) = 3$
3. $\log_{10}(100) = 2$

Exercise 6: Rewrite as Logarithms

Rewrite each exponential equation as a logarithm.

1. $2^6 = 64$
2. $3^4 = 81$
3. $10^5 = 100000$

Exercise 7: Solve the Equations

1. $2^x = 32$

2. $10^x = 1000$

3. $3^x = 81$

4. $\log_2(x) = 5$

5. $\log_{10}(x) = 4$

Exercise 8: Word Problems

1. A savings account contains €500 and grows by 5% every year.
 - (a) Write a formula for the balance after t years.
 - (b) How much money is in the account after 4 years?

2. A rumor spreads through a school. The number of students who know it triples every day. On the first day, 4 students know the rumor.
 - (a) Write an exponential model.
 - (b) How many students know the rumor after 5 days?

Challenge

Without using a calculator, solve:

$$2^{x+1} = 64.$$

$$\log_2(x - 1) = 3.$$

Explain in your own words why logarithms are called the *inverse* of exponential functions.

Exercise 9: The Natural Exponential Function

Recall that

$$\exp(x) = e^x.$$

Calculate the following. Give exact answers where possible.

1. $\exp(0) = \underline{\hspace{2cm}}$
2. $\exp(1) = \underline{\hspace{2cm}}$
3. $\exp(2) = \underline{\hspace{2cm}}$
4. $\exp(-1) \approx \underline{\hspace{2cm}}$

Rewrite each expression using powers of e .

1. $\exp(3)$
2. $\exp(-2)$
3. $\exp(x)$

Exercise 10: The Natural Logarithm

Recall that

$$\ln(x) = \log_e(x).$$

Calculate.

1. $\ln(e) = \underline{\hspace{2cm}}$
2. $\ln(e^3) = \underline{\hspace{2cm}}$
3. $\ln(1) = \underline{\hspace{2cm}}$
4. $\ln(e^{-2}) = \underline{\hspace{2cm}}$

Rewrite each equation.

1. $e^3 = 20.09 \dots$
2. $\ln(x) = 2$
3. $e^x = 7$

Exercise 11: Solving Equations

Solve the following.

1. $e^x = e^5$

2. $\ln(x) = 4$

3. $e^{2x} = e^8$

4. $\ln(e^{-5})$

5. $e^{\ln(7)}$

6. $\ln(e^{12})$

Exercise 12

Typical loudness levels are

Whisper	30 dB
Conversation	60 dB
Busy street	80 dB
Rock concert	110 dB

Answer the following.

1. How many times greater is the intensity of a conversation than a whisper?
2. How many times greater is the intensity of a busy street than a conversation?
3. How many times greater is the intensity of a rock concert than a whisper?
4. What is the sound pressure for a whisper?
5. What is the sound pressure for a conversation?
6. What is the sound pressure for a busy street?
7. What is the sound pressure for a rock concert?

Exercise 13

- What pressure amplitude corresponds to 0 dB? Does this mean there is no sound at all?
- What pressure amplitude corresponds to 80 dB?
- Assume one vacuum cleaner produces a sound with 70 dB. How loud will it be if we turn on a second vacuum cleaner? How about ten vacuum cleaners?

Exercise 14

Rewrite the formula for decibels with the natural logarithm instead of $\log_{10}$.

$$20 \log_{10}\left(\frac{p_1}{p_{\text{ref}}}\right) \text{dB} \quad (1)$$

Exercise 15 (Challenge)

Imagine you have a sound source that radiates sound equally in all directions. Standing 10 metres away from it, the loudness is 60 dB. How loud is it at 5 metres distance from the source? What about 20 metres?

Hint: It is easier to think about the intensity than about the pressure here. Remember: Intensity is energy per unit time per unit area. The sound source is transporting energy to its surroundings. The energy must be conserved. We can assume that the energy is distributed evenly on the surface of a sphere with a certain radius. What happens as we increase or decrease the radius of the sphere? How does this affect the intensity? Now plug this into the dB formula below. It is the same as for the powers, since power and intensity are proportional.

$$10 \log_{10}\left(\frac{I_1}{I_2}\right) \text{dB} \quad (2)$$